

Technologies for E-Commerce

CAN302

**Department of Communications and Networking
Xi'an Jiaotong-Liverpool University (XJTLU)**

Lecture 1 – Introduction to Web technology



@Front-end



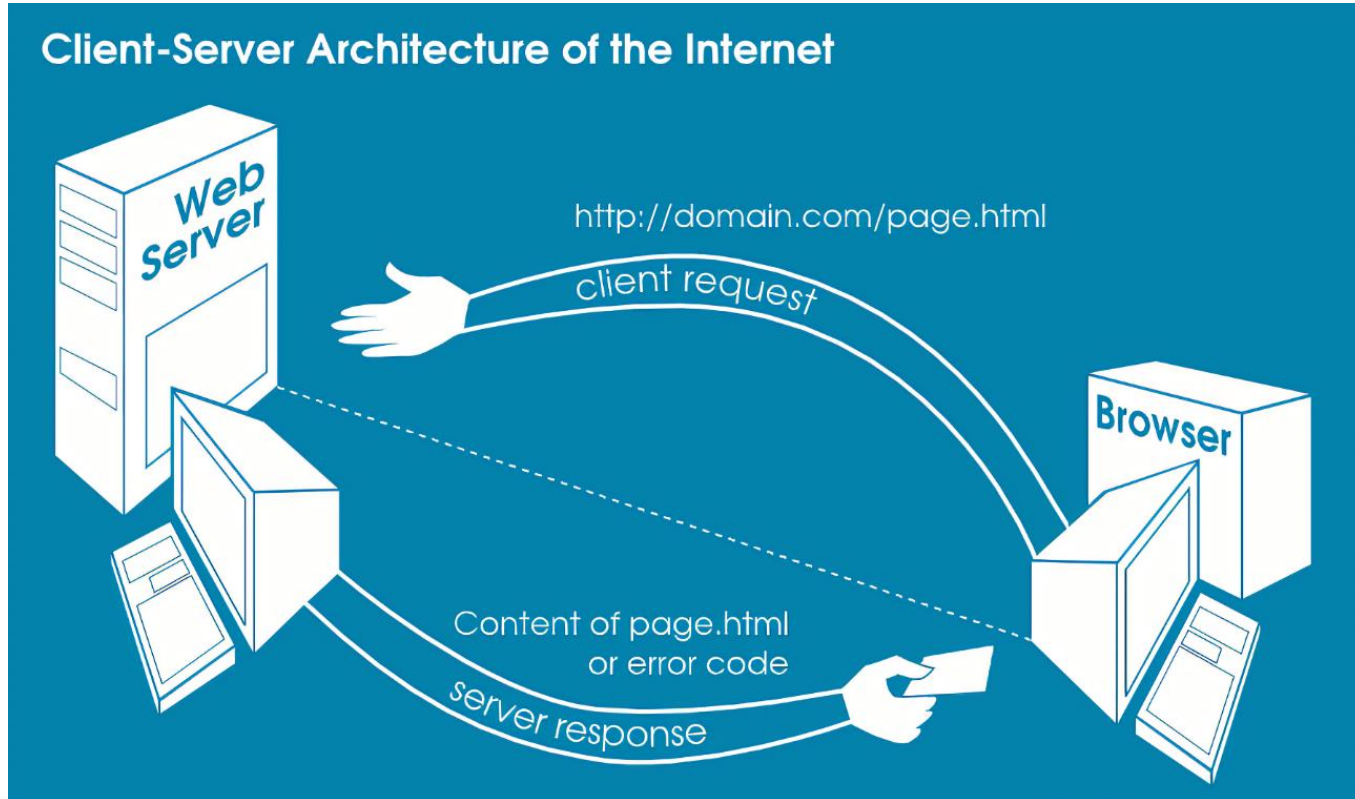
@Back-end

When we talk about e-commerce...



- **Buy something on-line**
- **Through what technologies? → APP @ mobile and WEB @ PC**

How does the webpage work?

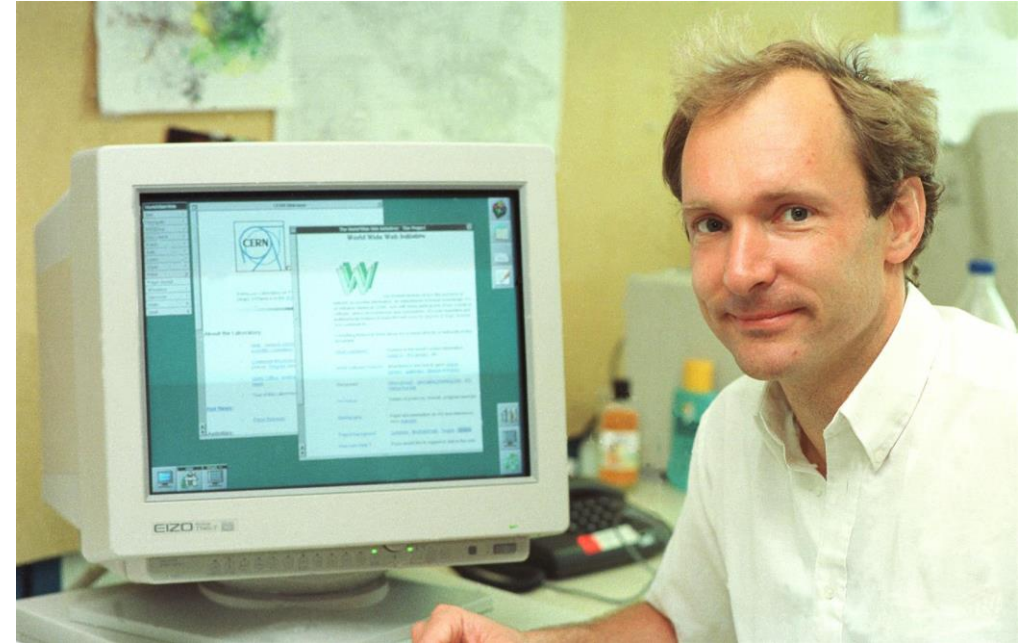


- Webpage works in a **Client-Server** mode, requested by the **Client** and response the request by the **Sever**.
- **HTTP** (hypertext transport protocol) is a protocol for fetching resources such as HTML documents.
- HTTP is the foundation of any data exchange of the Web applications (but **NOT** all network applications!).

Typical application protocols

Telnet	Allows users to access resources on another machine. All data is seen in clear text (not recommended for use)
Secure Shell (SSH)	Similar to Telnet but it sets up a secure session (recommended over telnet). All data is encrypted during the session
File Transfer Protocol (FTP)	Allows users to transfer files between two hosts. Users must use some type of Authentication, also includes advance file search features
Trivial File Transfer Protocol (TFTP)	Allows users to transfer files between two hosts. Users does not need Authentication, also you lose a lot of functions such as directory browsing abilities
Simple Network Management Protocol (SNMP)	Pulls data from networking devices to verify status information / error messages, etc.
Hyper Text Transfer Protocol (HTTP)	Allows you to display graphics, text, and links correctly. manages communication between web browsers and web servers
Hyper Text Transfer Protocol Secure (HTTPS)	Allows you to display graphics, text, and links correctly. Manages communication between web browsers and web servers. This version make sure your web communication is secure using SSL
Network Time Protocol (NTP)	Synchronize devices to ensure that all computer on the network has the correct time
Domain Name Services (DNS)	Resolves hostnames on a network
Dynamic Host Configuration Protocol (DHCP/BOOTP)	Assigns IP configurations to hosts on a network DHCP Conflicts: two hosts have the same IP information. If conflict is detected it is removed from the pool and the will not be assigned until the admin resolves the conflict by hand. APIPA - clients can automatically self-configure an IP address and subnet mask. This will allow communication if you have no DHCP available. Address: 169.254.0.1 - 169.254.255.254

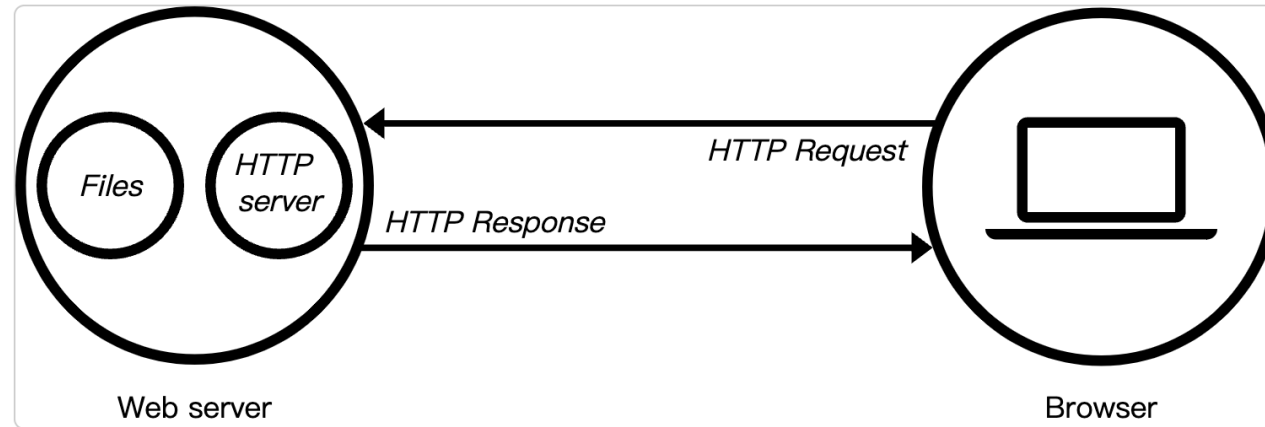
The invention of HTTP



- Tim Berners-Lee, a British scientist, invented the World Wide Web (WWW) in 1989, while working at CERN ("Conseil Européen pour la Recherche Nucléaire"). The web was originally conceived and developed to meet the demand for automated information-sharing.

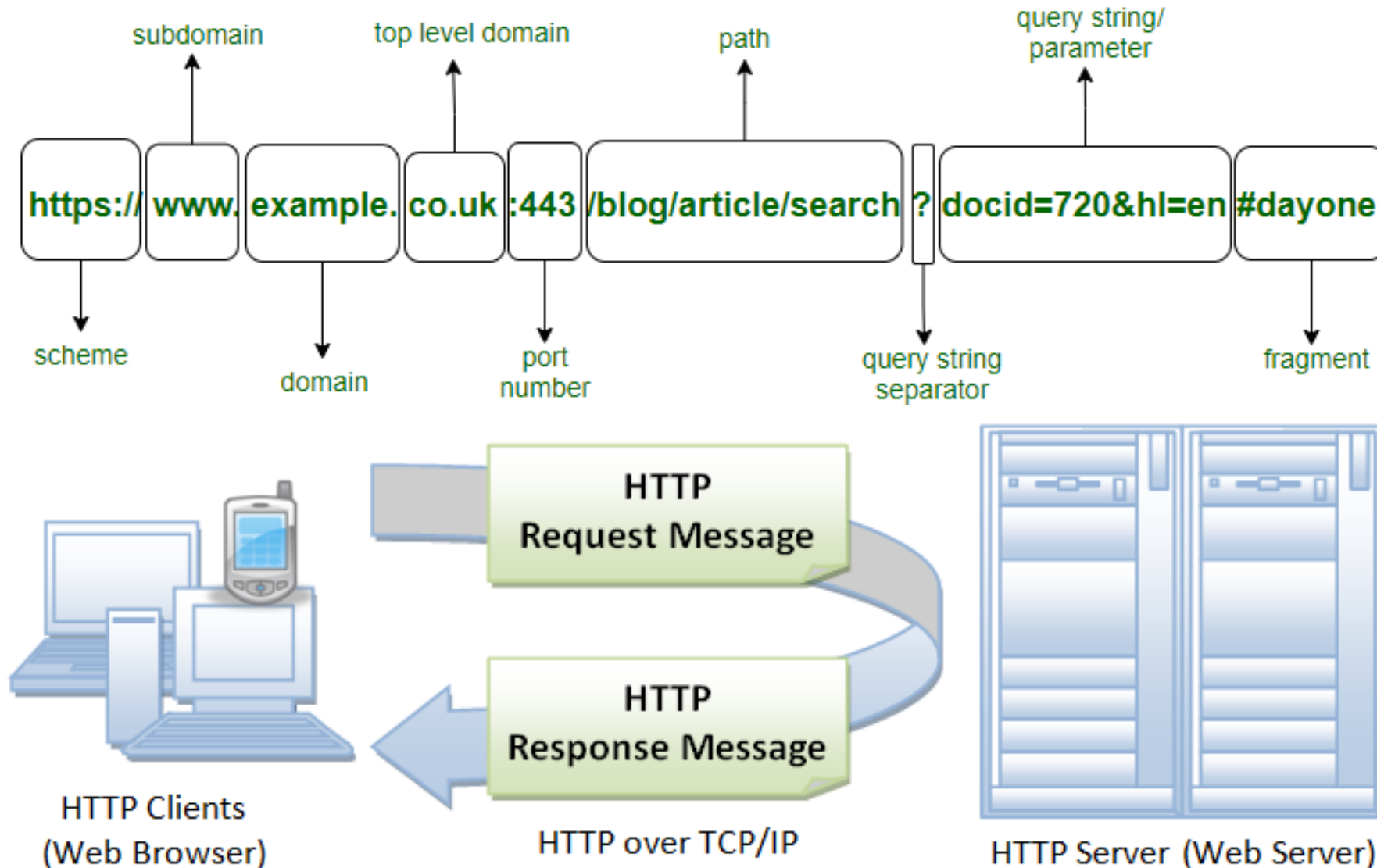
<https://info.cern.ch/>

HTTP – Hyper Text Transport Protocol

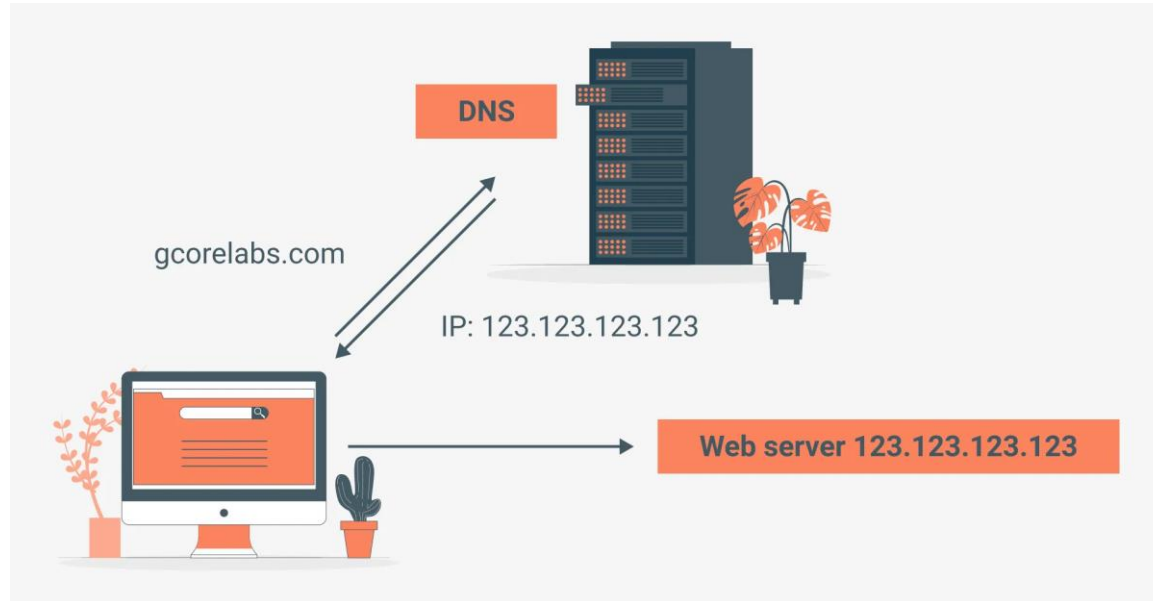


- Web server can refer to hardware or software, or both of them working together.
- On the hardware side, a web server is **JUST** a computer
- On the software side, a web server includes several parts that control how web users access hosted files. At a minimum, this is an HTTP server.
- An HTTP server is software that understands URL (Uniform Resource is the address of a unique resource on the internet) and HTTP (the protocol your browser uses to view webpages).
- Web Browsers are local applications acting as “client”
- They fetch data from web server, then render and show the contents to users.

What's in a web address?



Domain Name Service



- Translate the domain name to a IP address

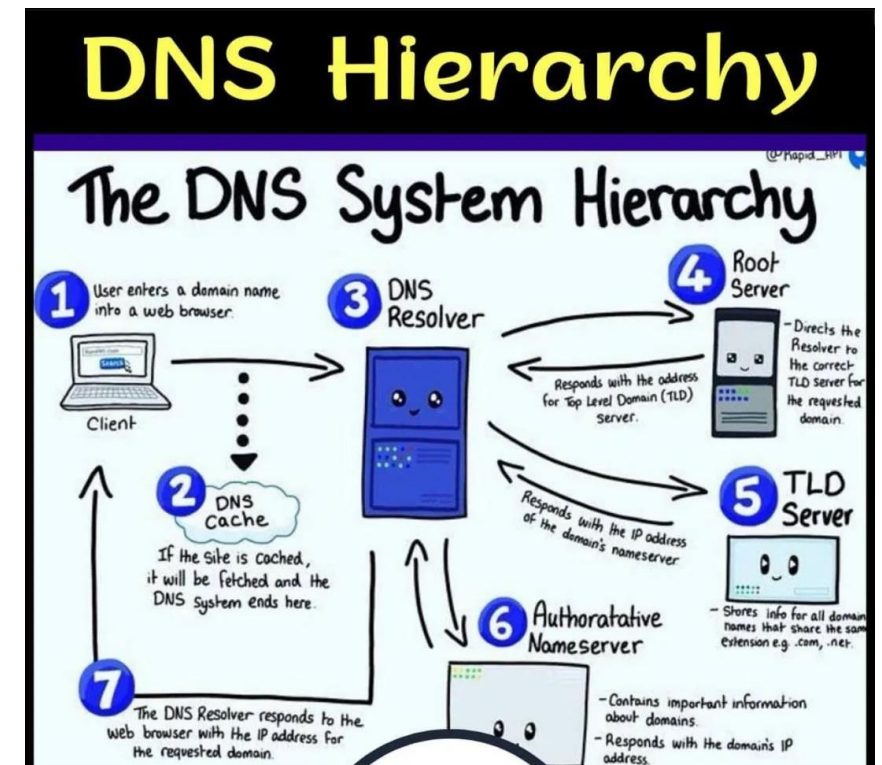
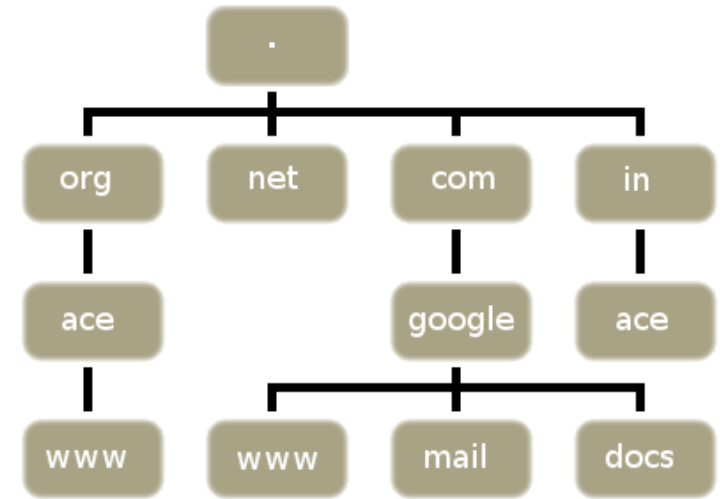
<https://docs.anushesh.in/linux/033-dnsservice.html>

Root Domain

Top Level Domains

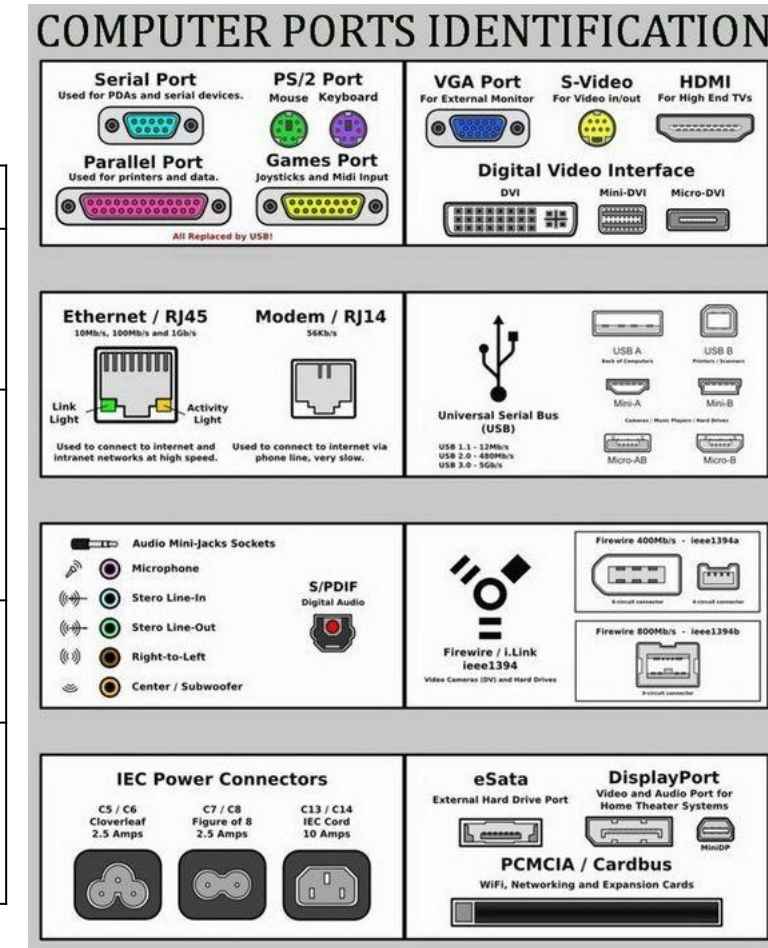
Second Level Domains

Third Level Domains



IP Address and ports

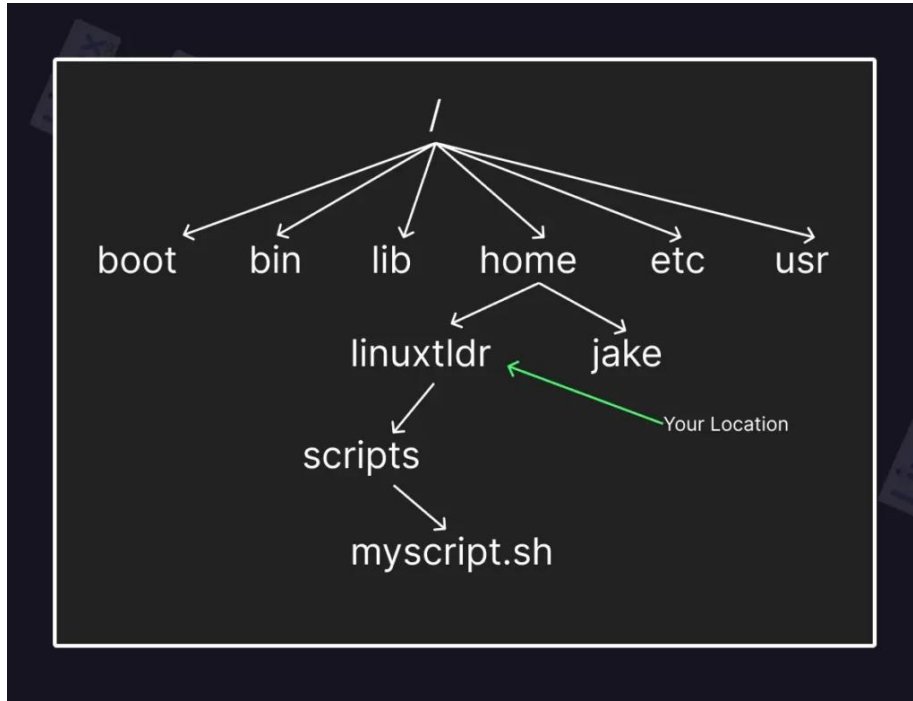
Feature	Port Number	IP Address
Purpose	Identifies a specific application or service on a device	Identifies a specific device on a network
Format	A number between 0 and 65535	A unique numerical label assigned to each device on a network
Example	Port 80 for HTTP, Port 443 for HTTPS	192.168.1.1, 255.255.255.0
Role in Networking	Directs incoming data to the correct application	Identifies the destination device for data packets



- Network ports are similar to hardware ports
- Browser and server need ports to communicate with each other

<https://www.revesoft.com/blog/telecom/port-number/>
<https://datacadamia.com/network/port>

IP Address and ports



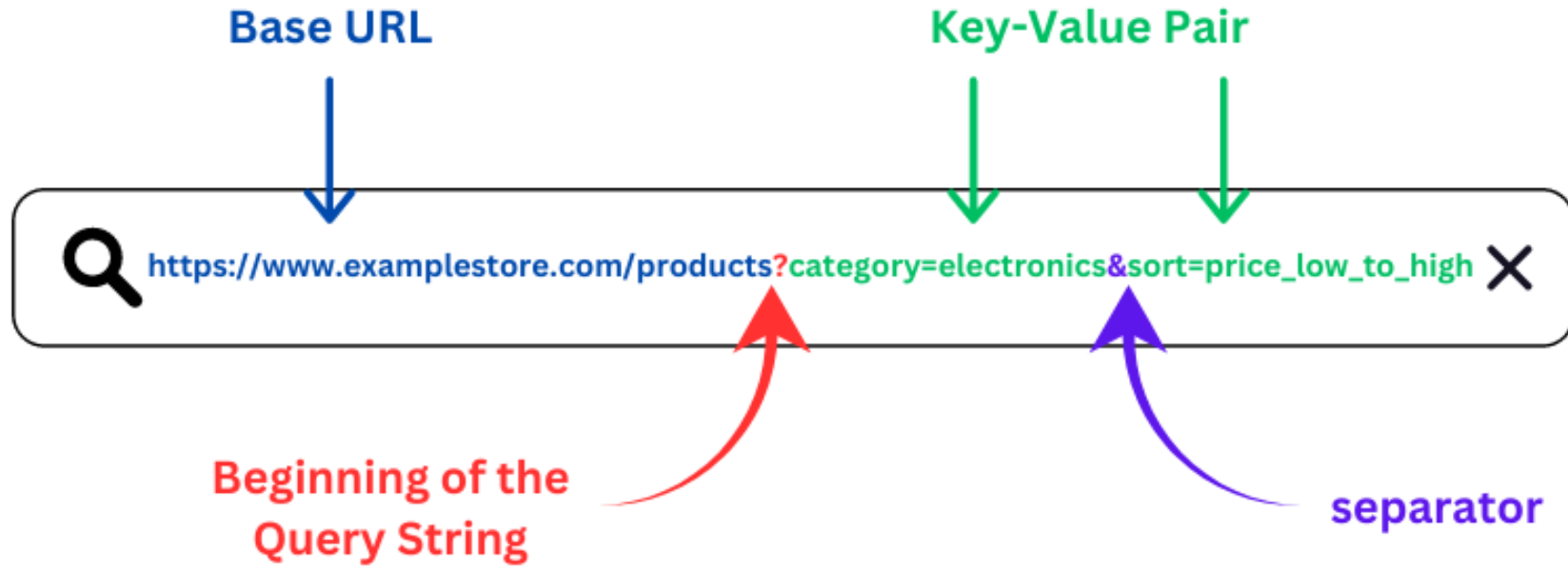
C:\Source\project\docs\file.txt

Root	Base	Relative	File
C:\	Source\project\docs\		file.txt

Working "Directory" Path Filename

- Computer files are stored in multi-level directories.
- Windows uses disk + path and Linux starts everything from "/" (root)
- Web server is a software running on the server computer. So it has their files on the server computer.
- For a simple web server, it has a default folder mapping with the root of URL. The path and file name map to the corresponding resources on the server computer.

Query string



- Web server receives the key value and make it available for server codes
- It is important way to pass parameters for **dynamic** web

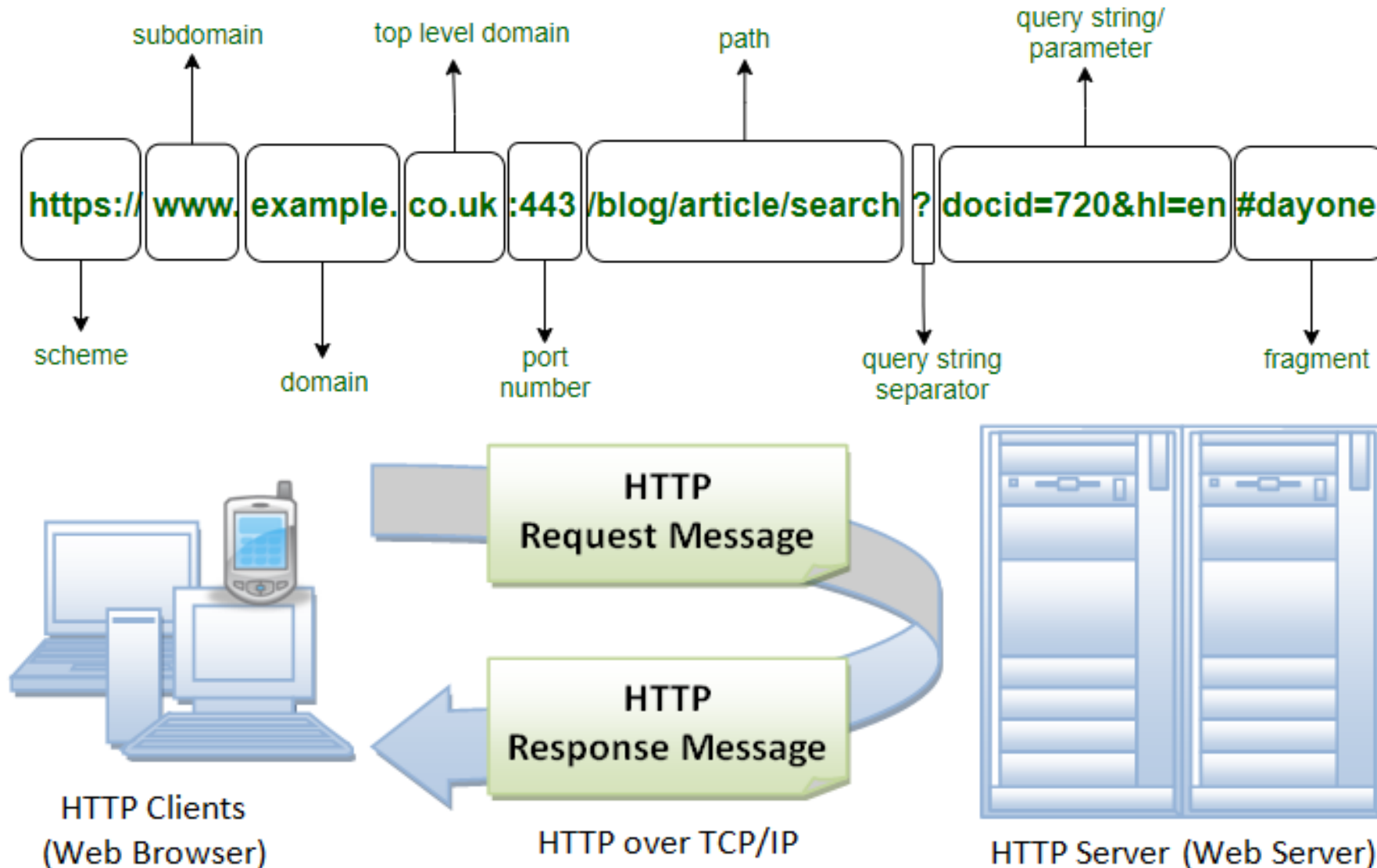
Document Fragment --

https://developer.mozilla.org/en-US/docs/Web/API/DocumentFragment/DocumentFragment#browser_compatibility

<https://developer.mozilla.org/en-US/docs/Web/API/DocumentFragment/DocumentFragment>

- The straightforward effect is auto-located to the corresponding part.

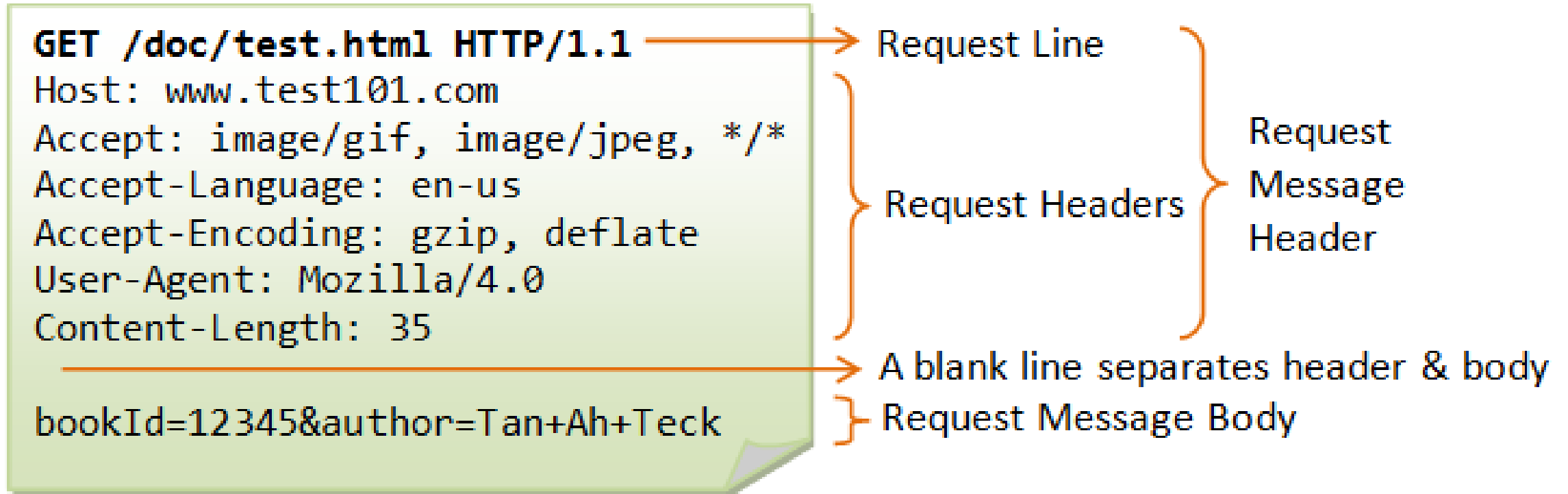
Ready to send a request



Http protocol – request methods

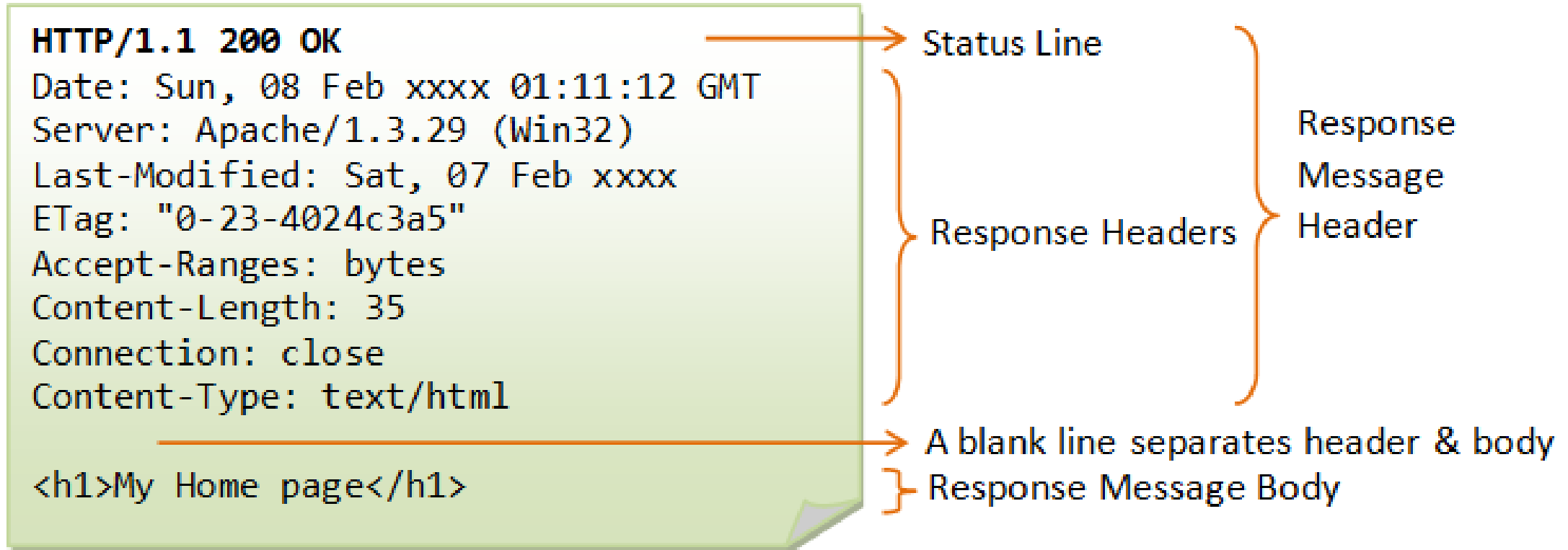
SAFE METHODS NO ACTION ON SERVER	{	GET	HTTP/1.1 MUST IMPLEMENT THIS METHOD
		HEAD	INSPECT RESOURCE HEADERS
MESSAGE WITH BODY SEND DATA TO SERVER	{	PUT	DEPOSIT DATA ON SERVER – INVERSE OF GET
		POST	SEND INPUT DATA FOR PROCESSING
		PATCH	PARTIALLY MODIFY A RESOURCE
		TRACE	ECHO BACK RECEIVED MESSAGE
		OPTIONS	SERVER CAPABILITIES
		DELETE	DELETE A RESOURCE – NOT GUARANTEED

The format of http request



- It can be found in Chrome browser Developer tools → network function.

The format of http response

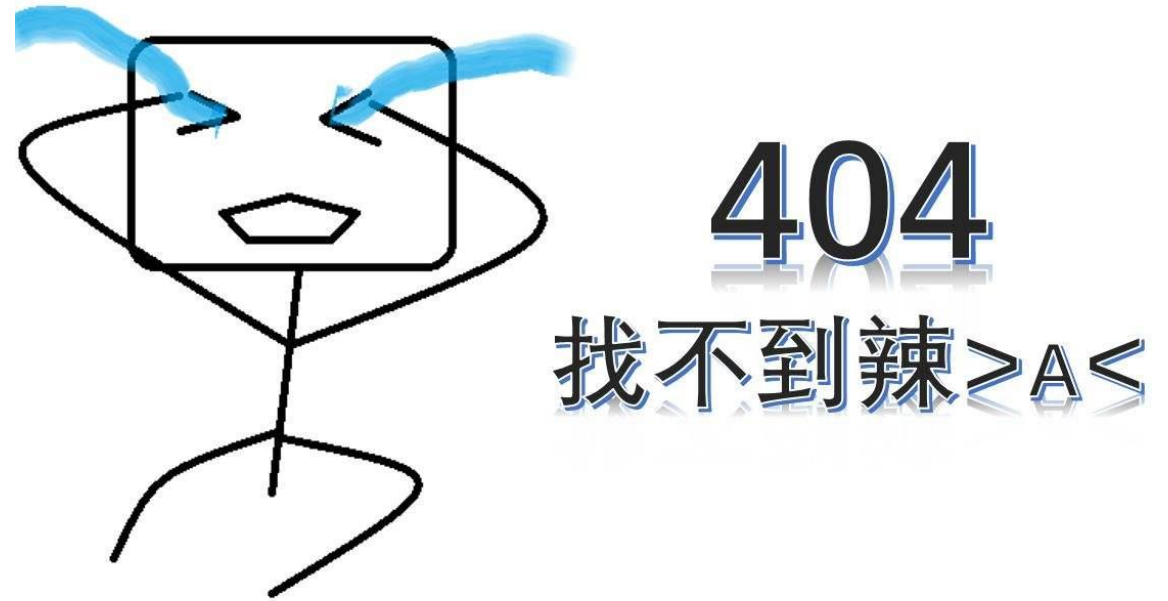


- It can be found in Chrome browser Developer tools → network function.

HTTP response code

HTTP response status codes indicate whether a specific [HTTP](#) request has been successfully completed. Responses are grouped in five classes:

- [Informational responses](#) (100–199)
- [Successful responses](#) (200–299)
- [Redirection messages](#) (300–399)
- [Client error responses](#) (400–499)
- [Server error responses](#) (500–599)



The status codes are defined by [section 10 of RFC 2616](#). You can find an updated specification in [RFC 7231](#).

<https://developer.mozilla.org/en-US/docs/Web/HTTP/Status>

What does Client and Serve do?



Front End

- Markup and web languages such as HTML, CSS and Javascript
- Asynchronous requests and Ajax
- Specialized web editing software
- Image editing
- Accessibility
- Cross-browser issues
- Search engine optimisation

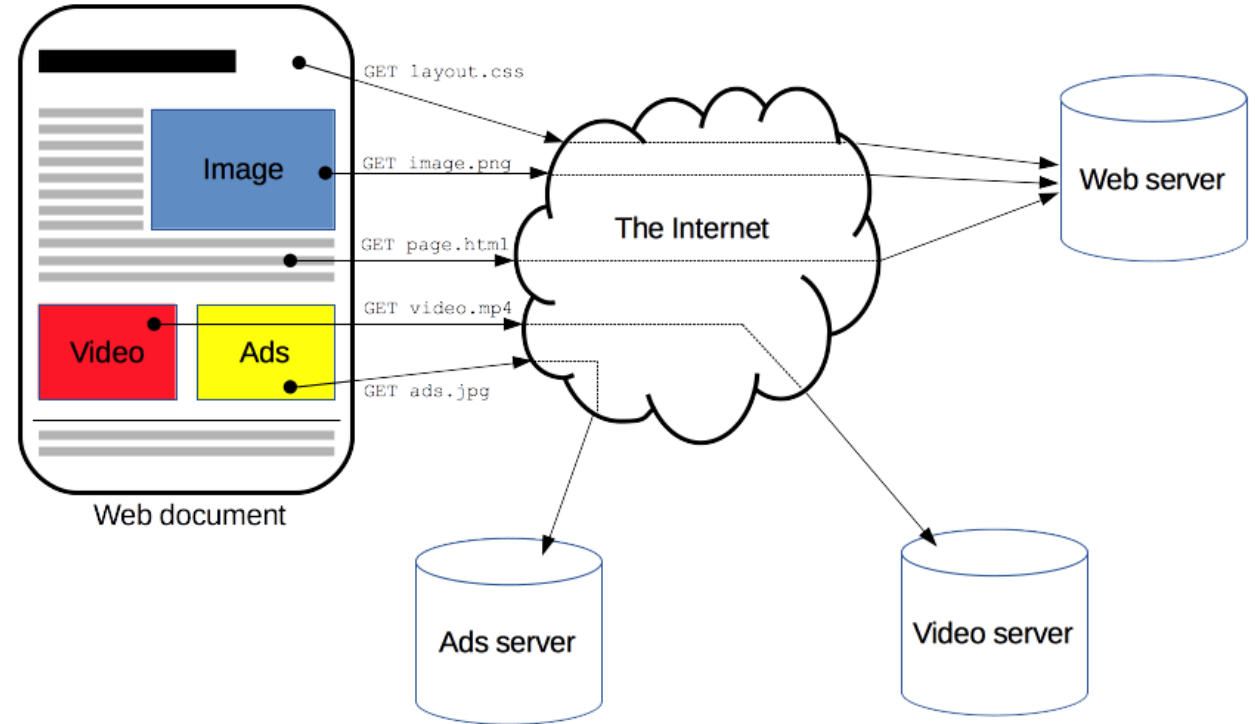
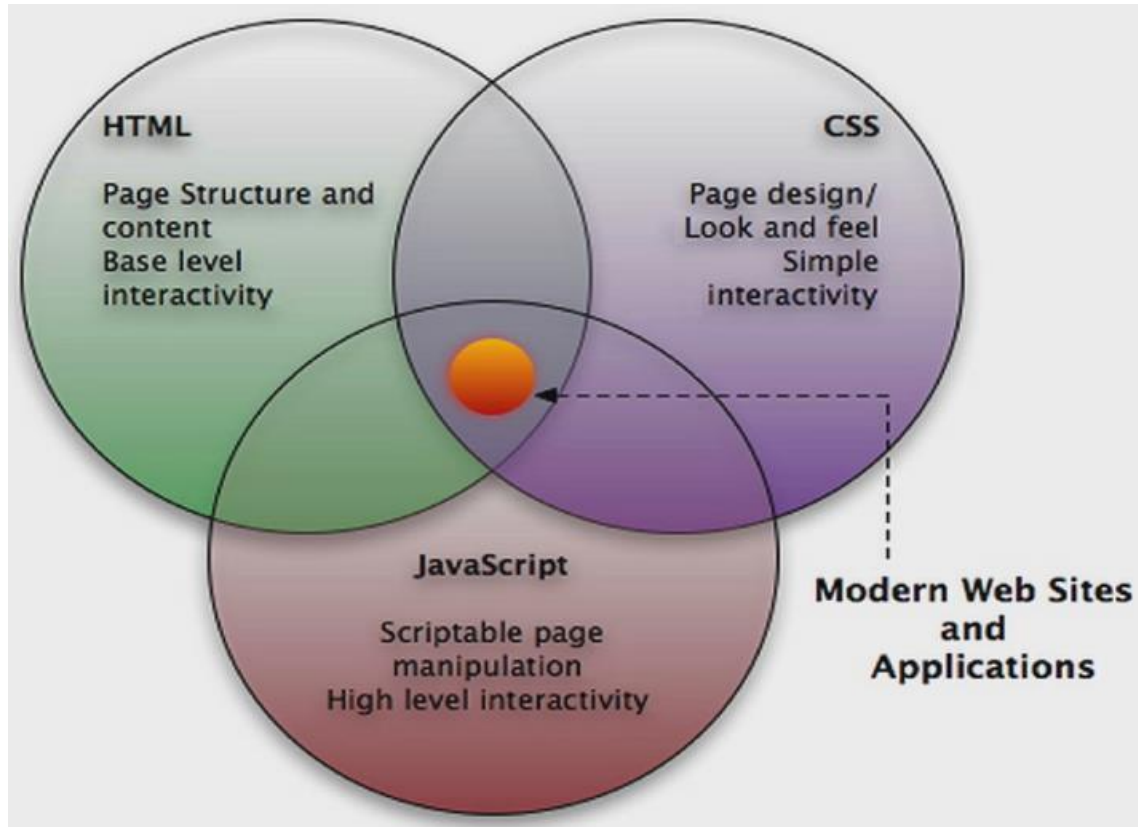


Back End

- Programming and scripting such as Python, Ruby and/or Perl
- Server architecture
- Database administration
- Scalability
- Security
- Data transformation
- Backup

- Back end refer to the server side. It **feed** the required data to front-end.
- Front-end refer to the web browsers. It requests the data from back-end and **show** them as a “web page” to end-user.
- A web project involves **at least FIVE languages**:
 - Front-end: HTML, CSS, JS
 - Back-end: one language and SQL

Web browser – Request and Render

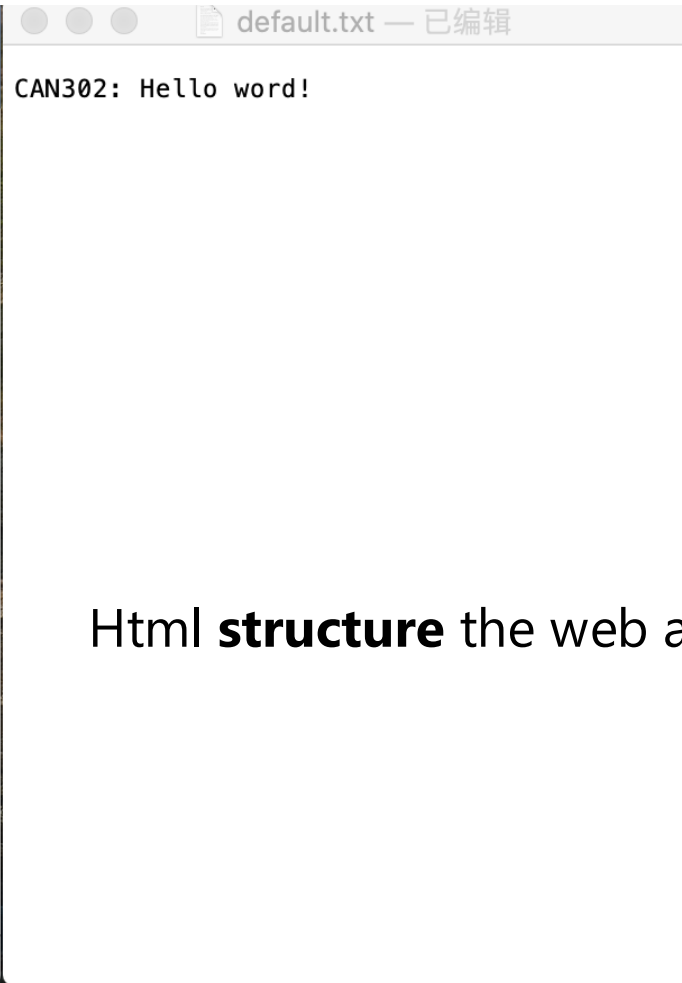


<https://html-css-js.com/>

<https://blog.hubspot.com/marketing/web-design-html-css-javascript>

<https://developer.51cto.com/art/202004/614896.htm>

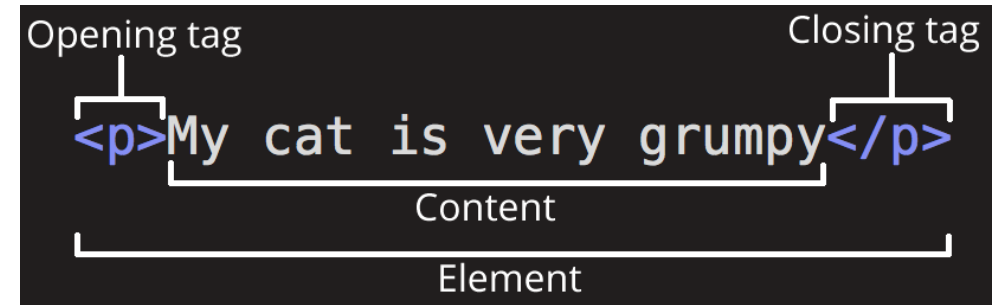
Think about “word” and “notepad”



Html **structure** the web and CSS give the **styles**!

Html basics

- Markup language
- Elements are the basic components in HTML
- Tags to mark the type of elements. Normally marks are pairs.
- Each element can set attributes. **Styles** define how the contents looks like. Styles can be defined directly or by a class.
- "id" is special attribute, it must be unique and can be referred as variable.



```
<p style="color:#ff0000;font-size:30px">这是第一段</p>  
<p style="color:#00ff00;font-size:40px">这是第二段</p>
```

```
<p id="p1"></p>  
<p id="p2"></p>  
<p id="p3"></p>
```

Html basics

```
<a href="https://www.mozilla.org/en-US/about/manifesto/">Mozilla Manifesto</a>
```

```
<!-- 4 heading levels: -->
<h1>My main title</h1>
<h2>My top level heading</h2>
<h3>My subheading</h3>
<h4>My sub-subheading</h4>
```

```
<p>At Mozilla, we're a global community of</p>

<ul>
  <li>technologists</li>
  <li>thinkers</li>
  <li>builders</li>
</ul>

<p>working together ... </p>
```

```
<p>This is a single paragraph</p>
```

HTML markup includes so many "elements" such as:

[<head>](#), [<title>](#), [<body>](#), [<header>](#), [<footer>](#), [<article>](#),
[<section>](#), [<p>](#), [<div>](#), [](#), [](#), [<aside>](#), [<audio>](#),
[<canvas>](#), [<datalist>](#), [<details>](#), [<embed>](#), [<nav>](#),
[<output>](#), [<progress>](#), [<video>](#), [](#), [](#), [](#) and
many others.

https://developer.mozilla.org/en-US/docs/Learn/Getting_started_with_the_web/HTML_basics

Html - Structuring the Web

```
<!DOCTYPE html>
<html>
  <head>
    <meta charset="utf-8">
    <title>My test page</title>
  </head>
  <body>
    
  </body>
</html>

<p>My cat is <strong>very</strong> grumpy.</p>
```

```

```

- You can put elements inside other elements too — this is called **nesting**. The whole html file is a nested by html, head and body and so on.
- Some elements have no content and are called empty elements. The sample has two attributes, but there is no closing `</img>` tag and no inner content.

CSS selector

Element Selector

```
h2 {  
  color: #c70039 ;  
}
```

Universal Selector

```
* {  
  color: #c70039 ;  
}
```

ID Selector

```
#content {  
  color: #6E4253;  
  font-size: 15px;  
}
```

Class Selector

```
.main {  
  margin-top: 10px  
  margin-bottom: 10px  
}
```

- A selector targets HTML to apply styles to content.
- It could be **HTML tag**, **id**, **Class** (start with ".") and more ...

<https://studyopedia.com/css/css-selectors/>

https://developer.mozilla.org/en-US/docs/Learn/CSS/First_steps/How_CSS_is_structured

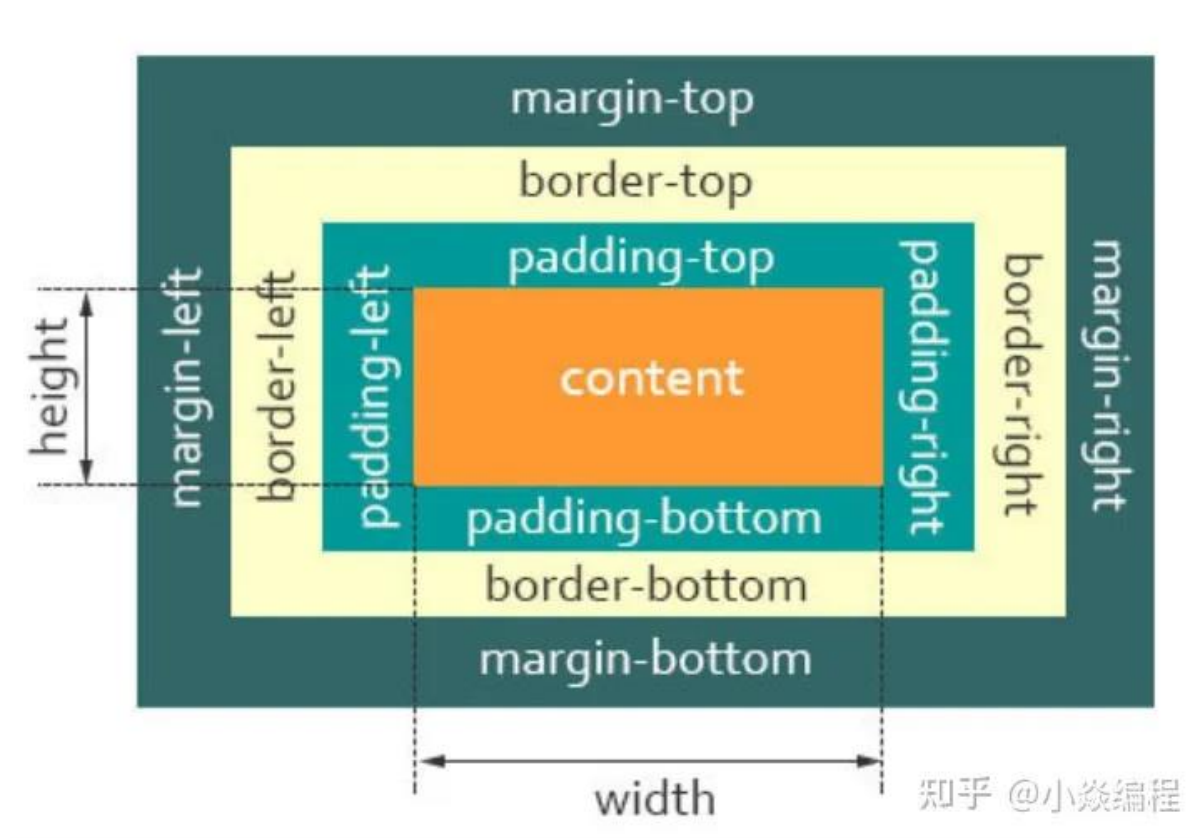
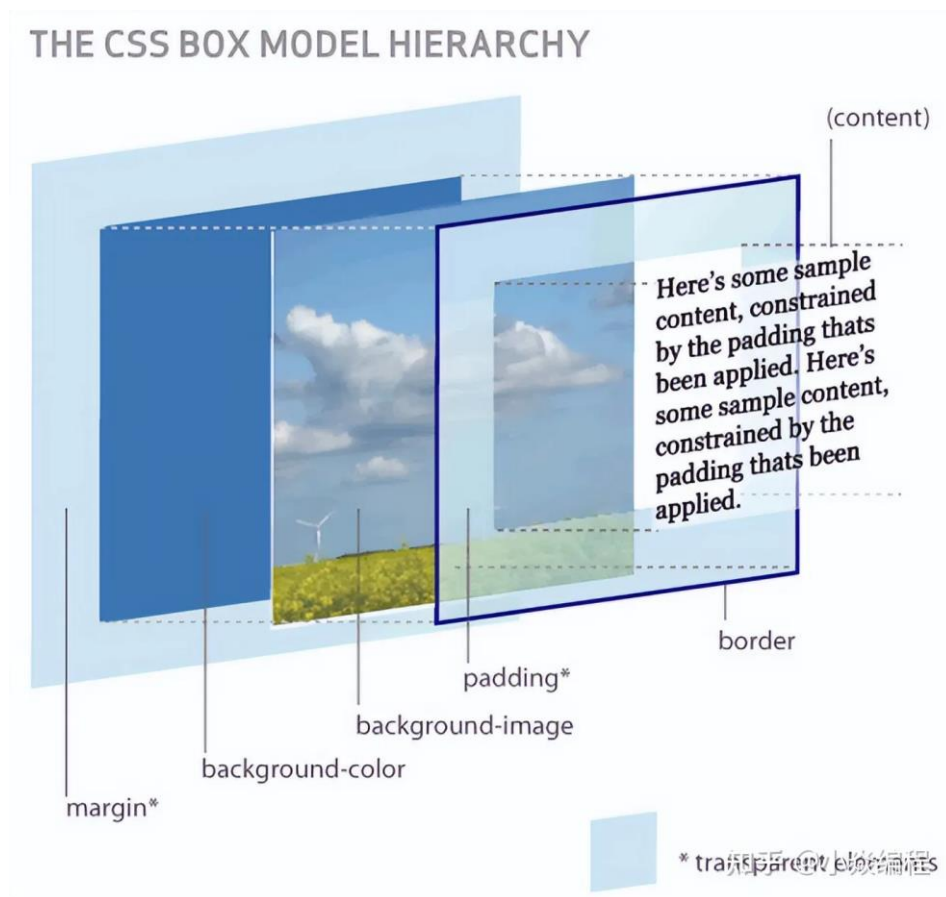
CSS property and value



- **Property and value:**
 - **Properties:** These are human-readable identifiers that indicate which stylistic features you want to modify. For example, [font-size](#), [width](#), [background-color](#).
 - **Values:** Each property is assigned a value. This value indicates how to style the property.

https://developer.mozilla.org/en-US/docs/Learn/CSS/First_steps/How_CSS_is_structured

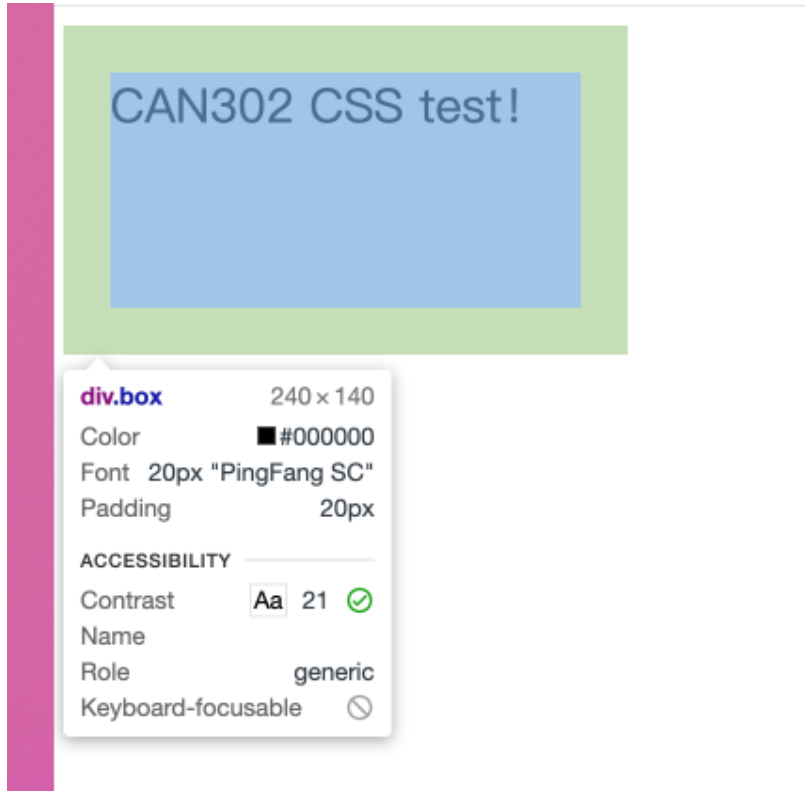
CSS – Box model



<https://zhuanlan.zhihu.com/p/546124925>

https://developer.mozilla.org/en-US/docs/Learn/CSS/Building_blocks/The_box_model

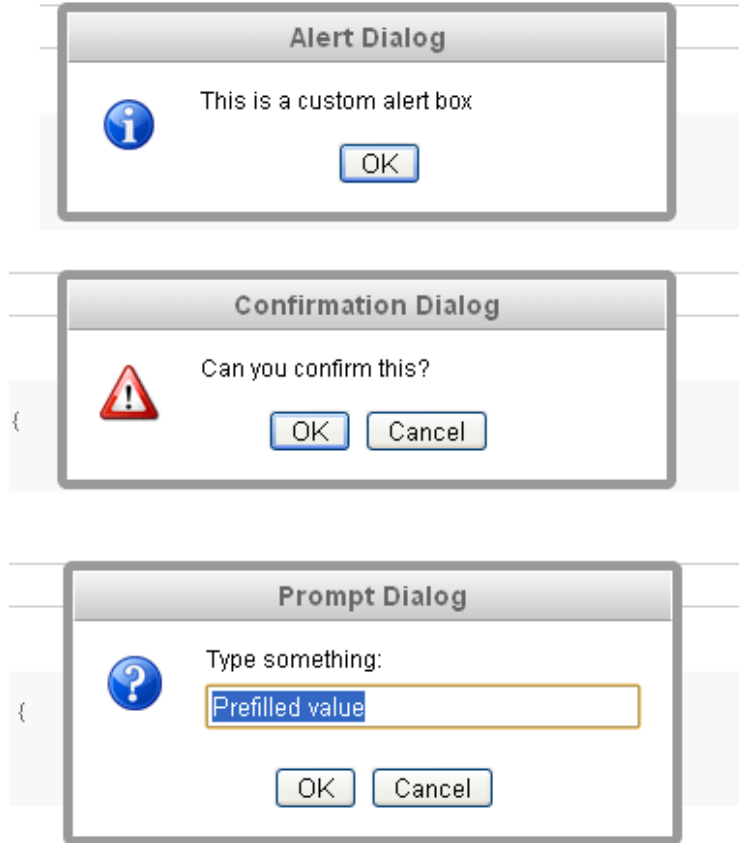
CSS – a box sample



```
1  <style>
2    .box {
3      width: 200px;
4      height: 100px;
5      padding: 20px;
6    }
7  </style>
8  <div class="box">
9    CAN302 CSS test!
10 </div>
```

- You can use browser to check the property of each element!

Mocha → LiveScript → JavaScript: JS

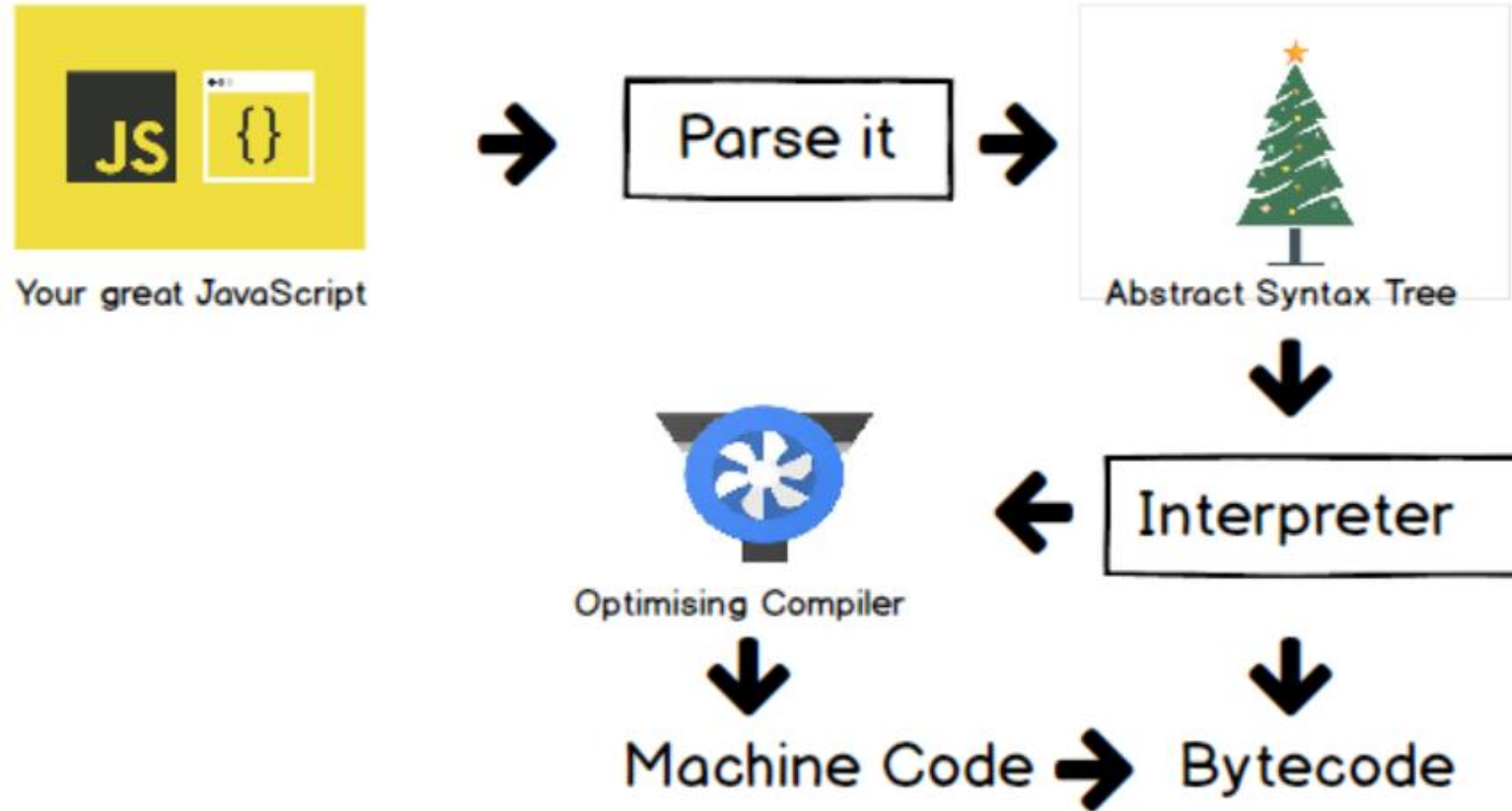


- Browser **has** JS engine which marry the best of an interpreter and a compiler.
- JS is a script language and TypeScript is a new choice for front-end.
- JS make the web interactive **without** server included.

Except **Java in **JavaScript**, JavaScript or JS does **NOT** behave the same with Java.**

<https://suleimanyunus.dev/how-javascript-works-the-javascript-engine>

Browser has the built in JS engine

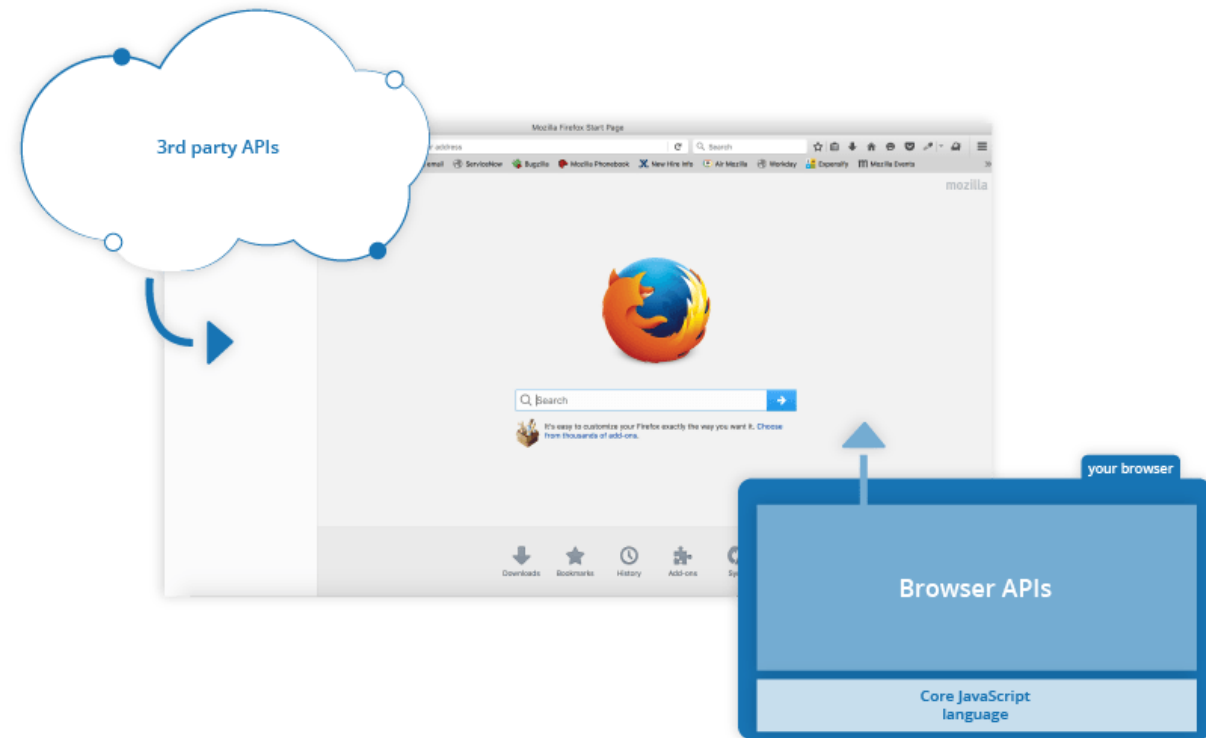


- **JS is a mixture of interpreted & compile language.**

<https://www.jianshu.com/p/c71c714c27bd>

<https://www.digitalocean.com/community/tutorials/js-v8-engine>

JS APIs - Application Programming Interfaces



- **APIs** provide you with extra superpowers to use in your JavaScript code.
- **APIs** are ready-made sets of code building blocks that allow a developer to implement programs that would otherwise be hard or impossible to implement.
- They do the same thing for programming that ready-made furniture kits do for home building — it is much easier to take ready-cut panels and screw them together to make a bookshelf.
- Browsers natively support APIs and there are 3rd party APIs.

https://developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps/What_is_JavaScript

A sample of Html, CSS and JS

Settings ⚙️ Include +

HTML

```
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <title>HTML CSS JS</title>
5 </head>
6 <body>
7 <h1 id="welcome">HTML CSS JS</h1>
8 <p>Welcome to HTML-CSS-JS.com</p>
9 <p>Online HTML, CSS and JavaScript editor
10 with instant preview.</p>
11 </body>
12 </html>
```

JavaScript

```
1 // JavaScript
2 document.getElementById('welcome').innerText +=
3 " CAN302 ";
4
```

CSS

```
1 /* CSS styles */
2 h1 {
3 font-family: Impact, sans-serif;
4 color: #CE5937;
5 }
6
```

HTML CSS JS CAN302

Welcome to HTML-CSS-JS.com

Online HTML, CSS and JavaScript editor with instant preview.

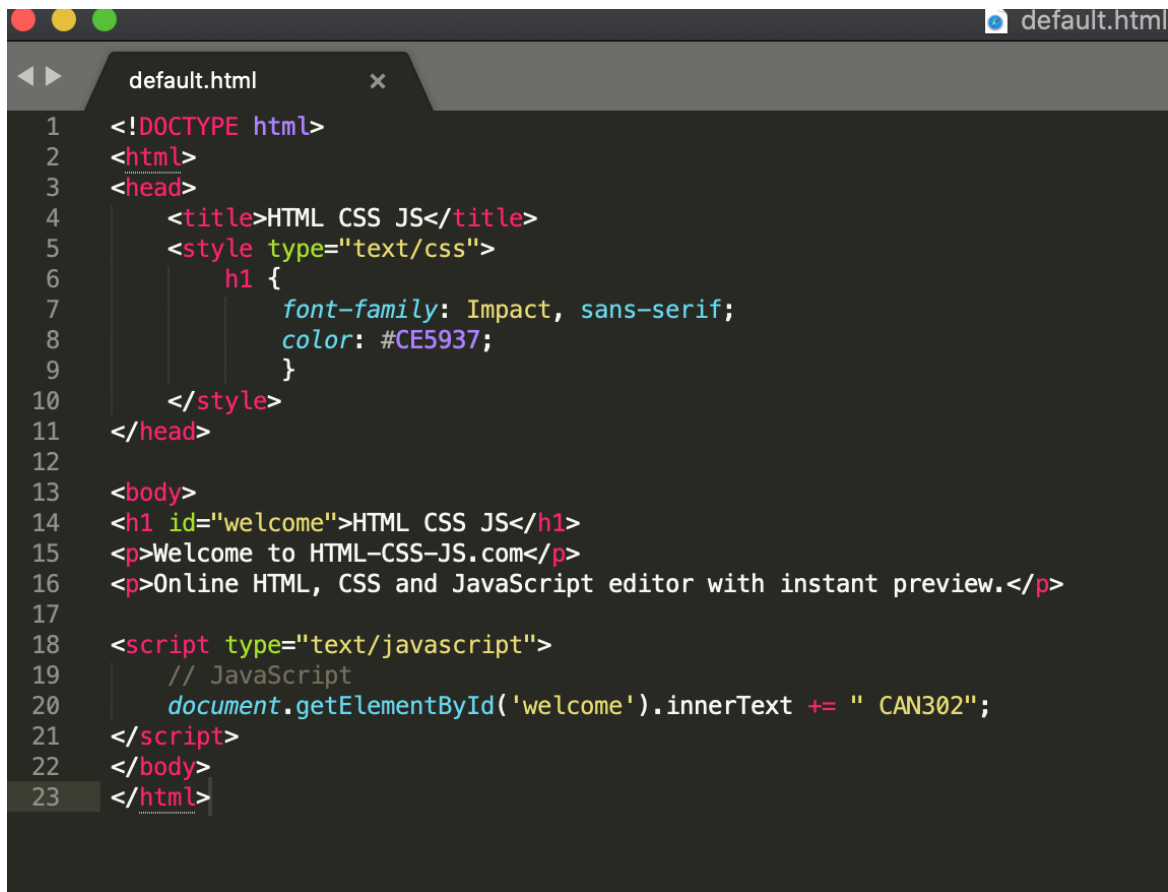
<https://html-css-js.com/>

USE IDE to edit the codes



- We can write any code in notepad but the efficiency would be very poor.
- IDE (Integrated development environment) will help us to have a better development experience.
- There are many choices, such as VS code, Sublime Text, WebStorm
- You can use any one your are familiar with.

All in one



```
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>HTML CSS JS</title>
5   <style type="text/css">
6     h1 {
7       font-family: Impact, sans-serif;
8       color: #CE5937;
9     }
10  </style>
11 </head>
12
13 <body>
14 <h1 id="welcome">HTML CSS JS</h1>
15 <p>Welcome to HTML-CSS-JS.com</p>
16 <p>Online HTML, CSS and JavaScript editor with instant preview.</p>
17
18 <script type="text/javascript">
19   // JavaScript
20   document.getElementById('welcome').innerText += " CAN302";
21 </script>
22 </body>
23 </html>
```



HTML CSS JS CAN302

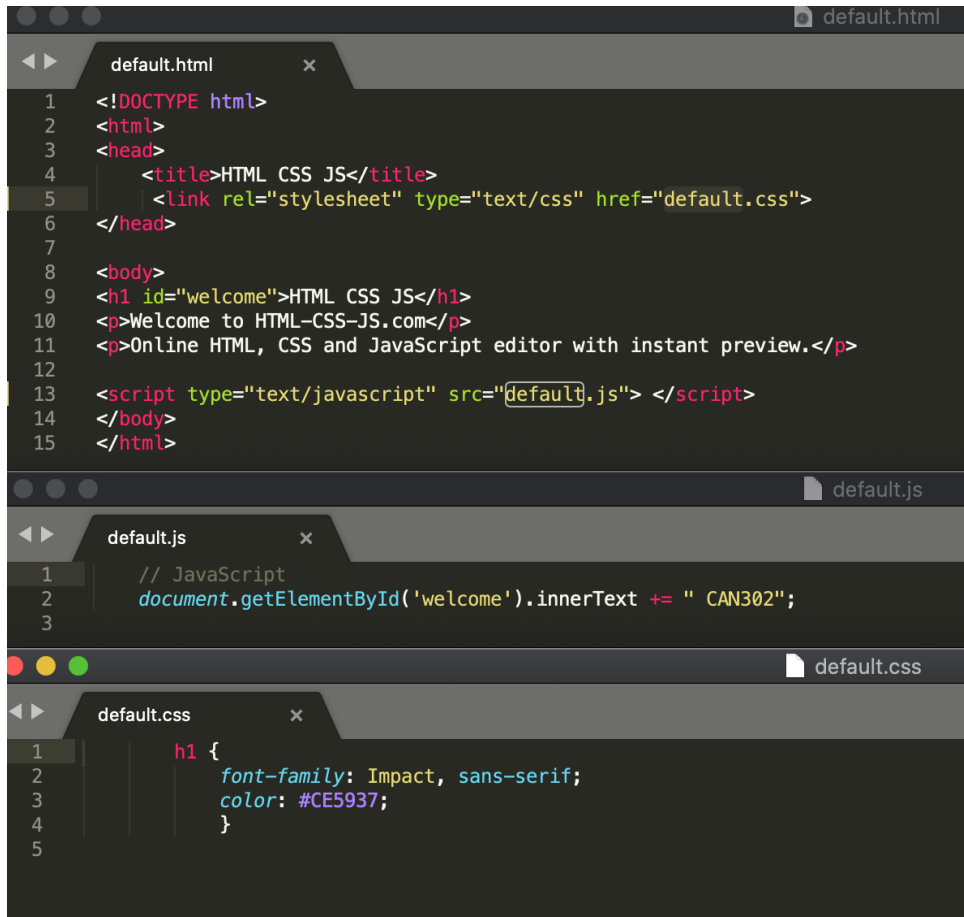
Welcome to HTML-CSS-JS.com

Online HTML, CSS and JavaScript editor with instant preview.

Browser can directly display the local html files

<https://www.cnblogs.com/hh-kk/p/7775261.html>

Save html CSS and JS in different files



The screenshot shows a code editor with three files open. The top file, 'default.html', contains HTML code with a title 'HTML CSS JS', a link to 'default.css', and a script tag for 'default.js'. The middle file, 'default.js', contains a JavaScript line to update the text of an element with ID 'welcome'. The bottom file, 'default.css', contains a CSS rule for the 'h1' element, setting the font-family to 'Impact, sans-serif' and the color to '#CE5937'.

```
default.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4   <title>HTML CSS JS</title>
5   <link rel="stylesheet" type="text/css" href="default.css">
6 </head>
7
8 <body>
9 <h1 id="welcome">HTML CSS JS</h1>
10 <p>Welcome to HTML-CSS-JS.com</p>
11 <p>Online HTML, CSS and JavaScript editor with instant preview.</p>
12
13 <script type="text/javascript" src="default.js"> </script>
14 </body>
15 </html>

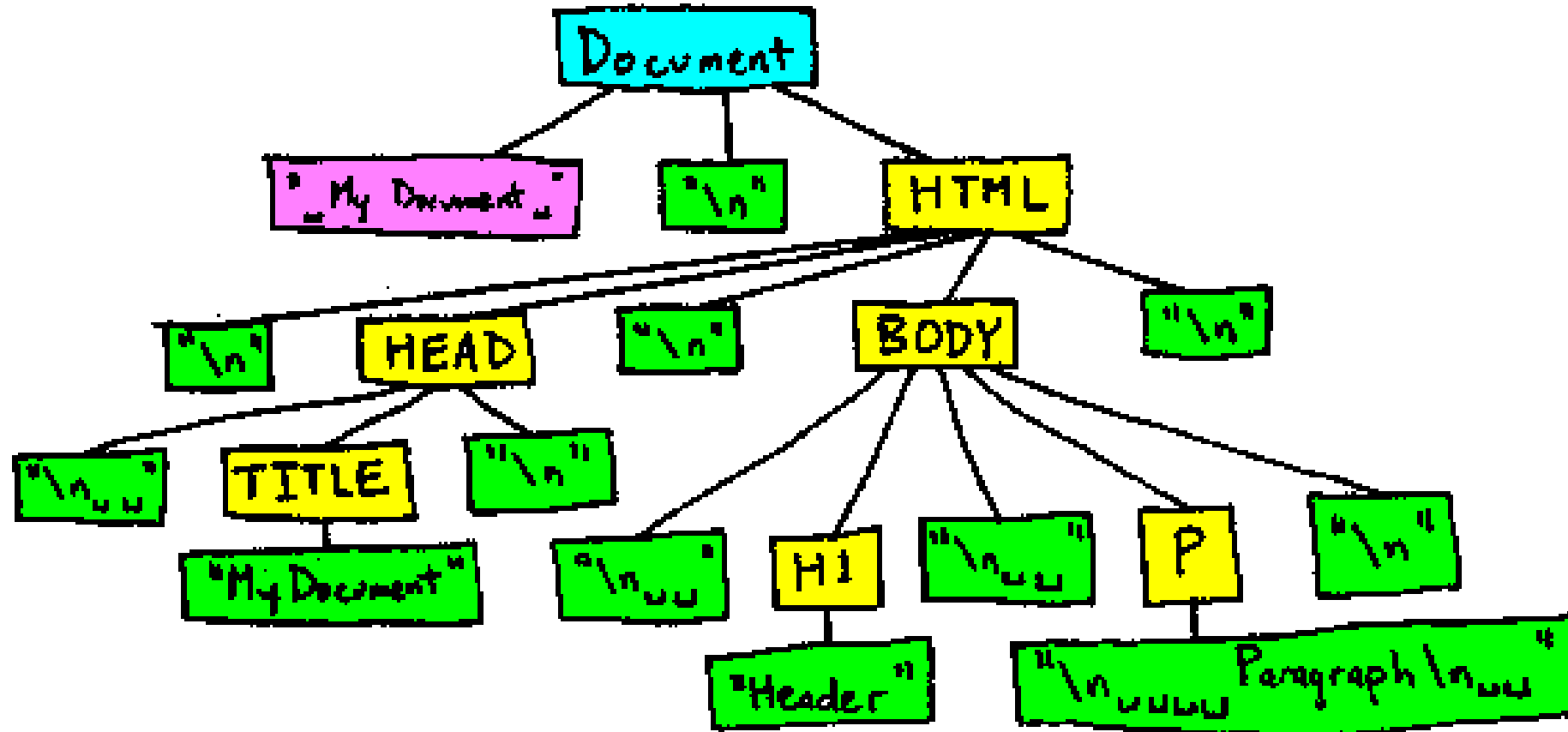
default.js
1 // JavaScript
2 document.getElementById('welcome').innerText += " CAN302";
3

default.css
1 h1 {
2   font-family: Impact, sans-serif;
3   color: #CE5937;
4 }
5
```

- A website always has much more than 1 page.
 - CSS and JS can be shared by different web pages.
 - We can put the CSS and JS in separate files.
 - We can use “tags” that bring in CSS and JS files.
-
- Same as all programming, please keep the **GOOD CODE MANNER**.

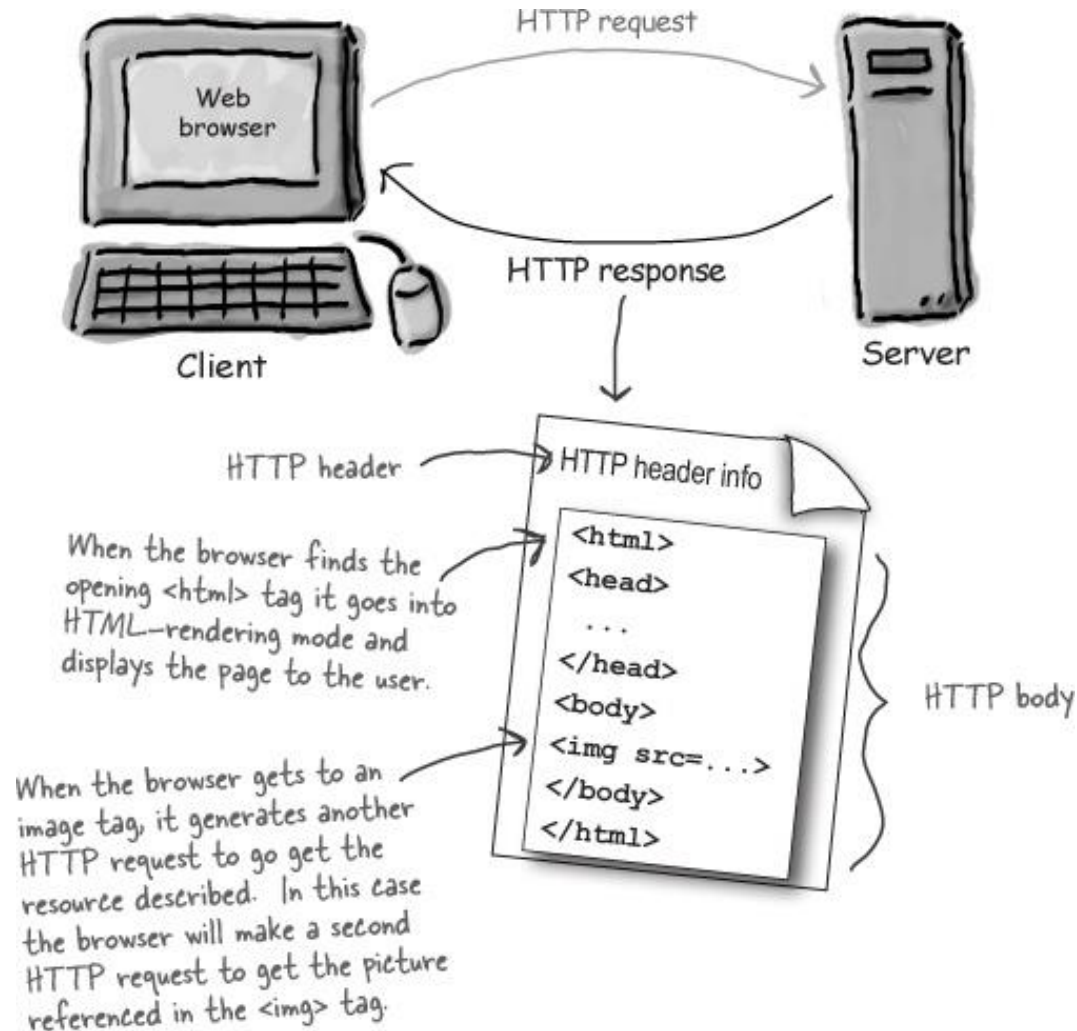
<https://www.bbsmax.com/A/QW5YZ7DGdm/>

Html CSS and JS together



[https://developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps/What is JavaScript](https://developer.mozilla.org/en-US/docs/Learn/JavaScript/First_steps/What_is_JavaScript)
[https://hurna.io/fr/academy/glossaries/web http.html](https://hurna.io/fr/academy/glossaries/web_http.html)

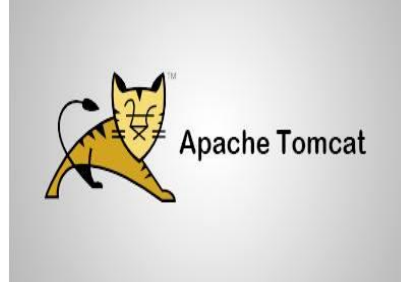
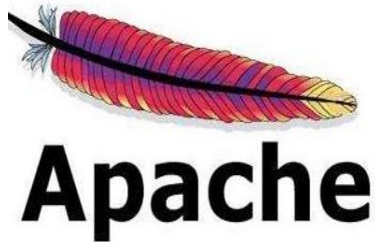
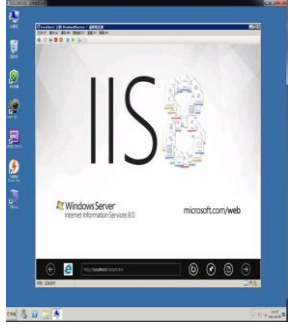
WEB server – store the content remotely



/usr/local/apache/htdocs/				
名字	大小	已改变	权限	拥有者
+		2019/4/3 星期三 21:18...	rw-r--r--	root
index.html	1 KB	2007/6/12 星期二 2:53...	rw-r--r--	root

- A basic HTTP server is a software that understands URLs (web addresses) and send the corresponding file stored on the server computer to client through HTTP response

Many choices for web servers

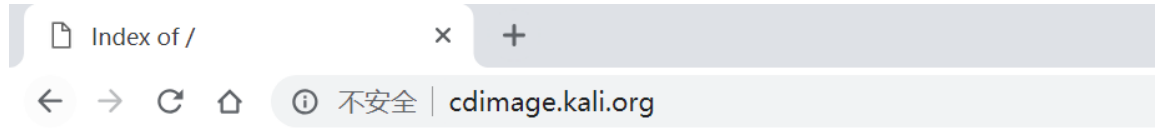


- There are many different web servers.
- Normally it would be optimized for a certain purpose and/or for a specific language.
- XAMPP is an easy to install **Apache server** distribution containing MariaDB(MySQL), PHP, and Perl. Which has a relatively low overall barrier for beginner.
- It used PHP, a language similar to Python.



- If you have XAMPP already, you can use it.
- If you have installed MySQL database, the XAMPP's MySQL cannot work due to port conflicts. You may uninstall the old MySQL.

List of all folders



Index of /

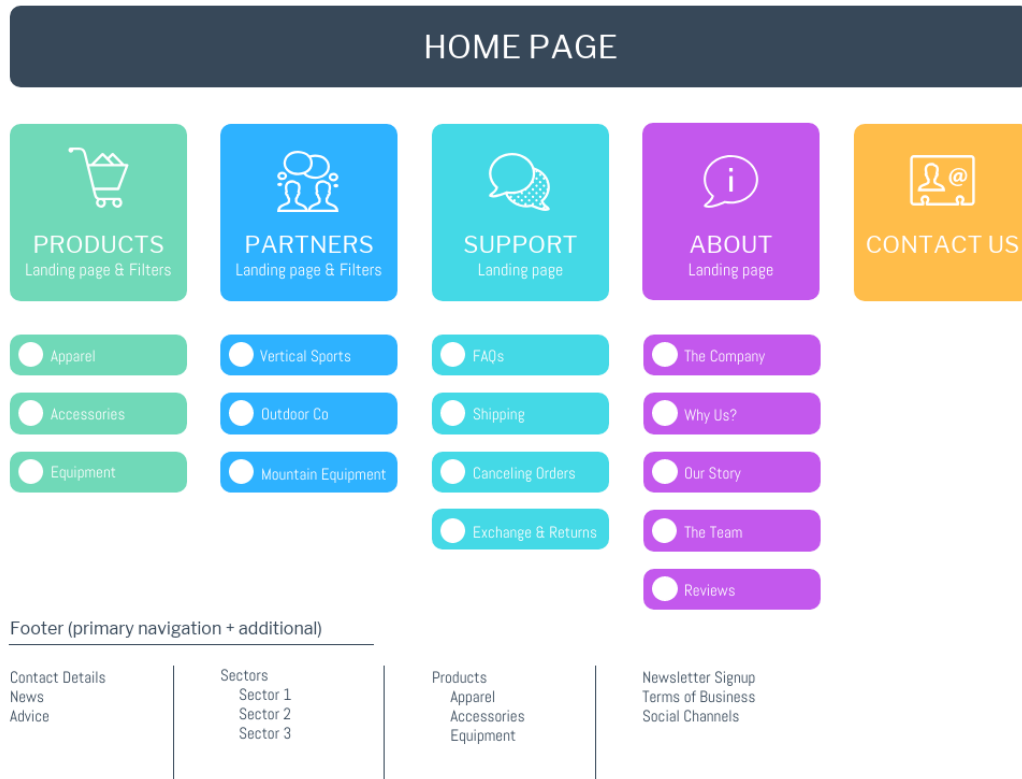
<u>Name</u>	<u>Last modified</u>	<u>Size</u>	<u>Description</u>
 README	2019-01-14 13:57	519	
 current/	2019-05-19 16:40	-	
 kali-2018.3a/	2018-09-14 19:33	-	
 kali-2018.4/	2018-10-29 07:16	-	
 kali-2019.1/	2019-02-17 19:06	-	
 kali-2019.1a/	2019-03-04 14:04	-	
 kali-2019.2/	2019-05-19 16:40	-	
 kali-weekly/	2019-07-21 04:03	-	
 project/	2014-12-04 14:11	-	

- Servers like Apache, it can allow to show all folders.
- Normally we want to **disable** this function since many "hidden" info will be leakage.
- You can see in this example, all web contents are corresponding to the files store in web server.

Apache/2.4.10 (Debian) Server at cdimage.kali.org Port 80

http://img.blog-photos.s3.amazonaws.com/361119192

How to create a website?



- **Write a html files for each page and Copy all files to the server computer!**

Each product for a HTML page ...

International Shopping: Shop E x Cloth Deals on eBay x +

amazon.com/b?node=16225009011&pf_rd_r=HV3Z0S5VDB26VZ87B413&pf_rd_p=5232c45b-5929-4ff0-8eae-5f67afd5c3dc&pd_rd_r=413e10b8-4e79-467d-92e5-013ed661db8a&pd_rd_w=BHgPy&pd_rd_w...

Department

- < Accessories & Supplies
- < Camera & Photo
- < Car & Vehicle Electronics
- < Cell Phones & Accessories
- < Computers & Accessories
- < GPS, Finders & Accessories
- < Home Audio
- < Office Electronics
- < Portable Audio & Video
- < Security & Surveillance
- < Television & Video
- < Video Game Consoles & Accessories
- < Video Projectors
- < Wearable Technology
- < eBook Readers & Accessories
- < Headphones, Earbuds & Accessories
- < Household Batteries, Chargers & Accessories
- < Power Accessories
- < Tablets

More Sustainable Products

- ☐ Climate Pledge Friendly

Payment Plan

- ☐ Layaway Eligible

Amazon Certified

- ☐ Auto Replenishment
- ☐ Works with Alexa

From Our Brands

- ☐ Amazon Brands

Featured Brands

- ☐ Amazon Basics
- ☐ Apple
- ☐ Allun
- ☐ HP
- ☐ Energizer
- ☐ Acoyer
- ☐ Syntech

Packaging Option

Electronics

Shop home entertainment, TVs, home audio, headphones, cameras, accessories and more

1-12 of over 70,000 results for **Electronics**

Amazon's Choice



Apple Pencil (2nd Generation)
4.8 ★★★★★ ~ (73,052)
\$89.99 \$129.00
More Buying Choices
\$78.96 (13 used & new offers)

Best Seller



Amazon Basics 48 Pack AA High-Performance Alkaline Batteries, 10-Year Shelf Life, Easy to Open Value Pack
48 Count (Pack of 1)
4.7 ★★★★★ ~ (677,349)
Amazon brand

Best Seller



Amazon Basics 36 Pack AAA High-Performance Alkaline Batteries, 10-Year Shelf Life, Easy to Open Value Pack
36 Count (Pack of 1)
4.7 ★★★★★ ~ (647,524)
Amazon brand



Apple EarPods Headphones with Lightning Connector. Microphone with Built-in Remote to Control Music, Phone Calls, and Volume. Wired Earbuds for...
4.6 ★★★★★ ~ (191,362)

Best Seller

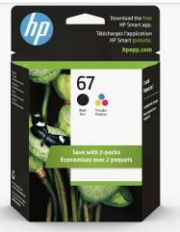


Apple 20W USB-C Power Adapter - iPhone Charger with Fast Charging Capability, Type C Wall Charger
4.8 ★★★★★ ~ (173,074)

Best Seller



Original HP 67XL Black High-yield Ink Cartridge | Works with HP DeskJet 1255, 2700, 4100 Series, HP ENVY 6000, 6400 Series | Eligible for Instant Ink |...
4.7 ★★★★★ ~ (33,388)



Original HP 67 Black/Tri-color Ink Cartridges (2 Count - Pack of 1) | Works with HP DeskJet 1255, 2700, 4100 Series, HP ENVY 6000, 6400 Series |...
4.7 ★★★★★ ~ (36,010)



Energizer AA Batteries, Double A Long-Lasting Alkaline Power Batteries, 32 Count (Pack of 1)
32 Count (Pack of 1)
4.8 ★★★★★ ~ (15,125)

Best Seller



Seagate Portable 2TB External Hard Drive HDD — USB 3.0 for PC, Mac, PlayStation, & Xbox -1-Year Rescue Service (STGX2000400)
4.7 ★★★★★ ~ (937,670)



Beats Solo3 Wireless On-Ear Headphones - Apple W1 Headphone Chip, Class 1 Bluetooth, 40 Hours of Listening Time, Built-in Microphone...
4.7 ★★★★★ ~ (61,541)



Roku Streaming Stick 4K | Streaming Device 4K/HDR/Dolby Vision with Roku Voice Remote and TV Controls
4.7 ★★★★★ ~ (50,112)
\$39.99 \$49.99

Best Seller



Amazon Basics 8 Units Pack 9 Volt Performance All-Purpose Alkaline Batteries, 5-Year Shelf Life, Easy to...
8 Count (Pack of 1)
4.7 ★★★★★ ~ (143,090)

- Manually updating the html pages would be a nightmare!

Dynamic webpage: frame + data



- Like a supermarket, we need to put products on the shelf in e-market.
- Manually update the products is not a possible way.
- It's time for dynamic web:

A dynamic website is the webpage on the server side where different contents are shown when assessed at different timings.

<https://www.educba.com/dynamic-website/>

<https://blog.csdn.net/dhp1027/article/details/7725653>

PHP - Hypertext Preprocessor

```
<html>
<body>

<?php
for ($i=1; $i<=5; $i++)
{
    echo "The number is " . $i . "<br />";
}
?>

</body>
</html>
```

Output:

```
The number is 1
The number is 2
The number is 3
The number is 4
The number is 5
```

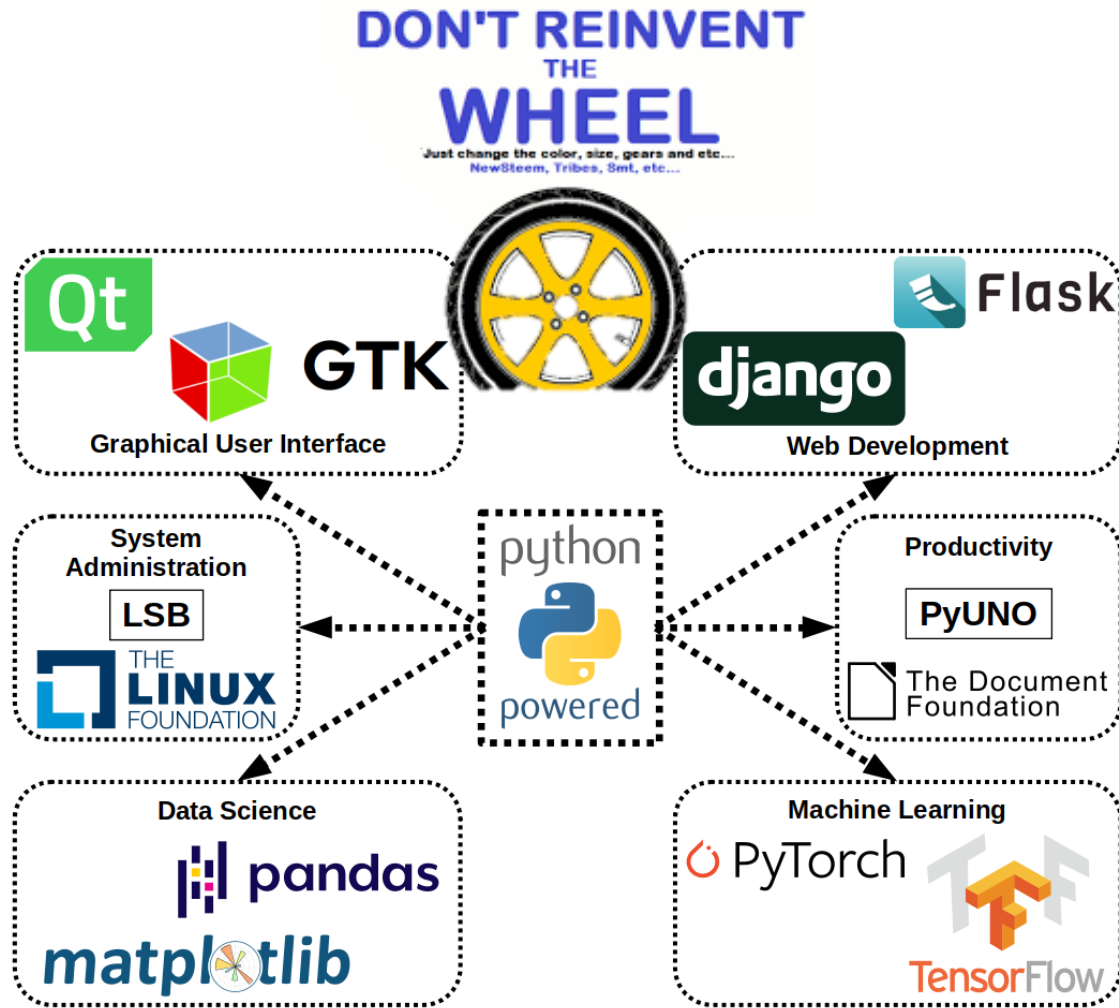
- PHP is a popular general-purpose scripting language that is especially suited to web development.
- The PHP scripts are embedded in html codes with opening tag `<?php` and closing tag `?>`
- When a PHP page is requested by the client, Web server (like Apache) will interpret the PHP scripts and use the output to replace the PHP scripts, the dynamic generated HTML content will be returned to the client.

Database basics



- Similar to **web server**, **database** is a software.
- A database is an organized collection of structured information, or data, typically stored electronically in a computer system.
- A database is usually controlled by a [database management system \(DBMS\)](#). Together, the data and the **DBMS**, along with the applications that are associated with them, are referred to as a database system.
- We often call a certain database which is actually the name of DBMS.

Software Framework



- Python was very popular because it got so many **"wheels"** already
- The designers of **software frameworks** aim to facilitate software developments by allowing designers and programmers to devote their time to meeting software requirements rather than dealing with the more standard low-level details of providing a working system, thereby reducing overall development time.

Bootstrap



- Quickly design and customize responsive mobile-first sites with Bootstrap, the world's most popular front-end **open source toolkit**, featuring Sass variables and mixins, responsive grid system, extensive prebuilt components, and powerful JavaScript plugins.
- Many **pre-defined classes** in Bootstrap includes CSS, **fonts** and JS (earlier version use jQuery).
- With Bootstrap, develop the front-end project can be much easier.
- We will explore more details in next Lecture!

<https://getbootstrap.com/>

LMO by Moodle

Learning Mall Core

Welcome to XJTLU online learning and teaching system!

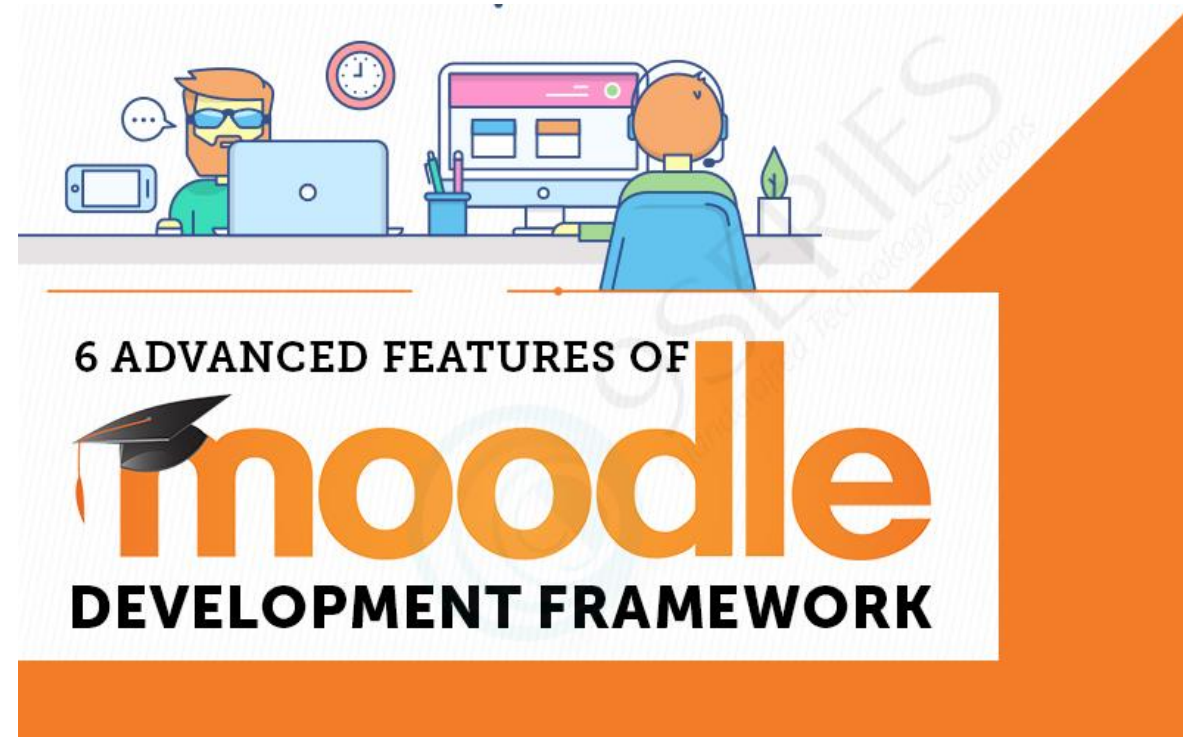
Please log in to access your XJTLU account and degree courses:

XJTLU Account

If you are a non-XJTLU user (without an XJTLU email address), please log in here:

NON-XJTLU Account

If you don't have the account above, you will need to create a new account by clicking [HERE](#). Please note: You need to use an enrolment key given by XJTLU staff to be authorised to create the new account.



- XJTLU's LMO is powered by Moodle

Summary

- Understand the client/server
- Understand HTTP
- Understand the mechanism of web browser
- Understand the mechanism of web server